

INDICE

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Big popipopi di inglese
(James Joyce nel programma non c'è ma io lo metto lo stesso)

Curriculum Vitae

Curriculum Vitae

What is

A CV is a formal document that provides an overview of your professional and educational history. Its main goal is to convince an employer that you are worth interviewing.

It should be clear, concise, and has to fit the job that you're applying for.

It should not be too long. The ideal length is about 2 pages.

It is also important to use a professional font, clear headings and even bullet points to make it easy to read.

What a CV should contains

A CV should contains:

- **Contact information**, such as your full name, email address, phone number and even linkedin profile or professional portfolio;
- **Professional Summary/Profile**: a brief (small) paragraph where you highlight your main strengths, who you are and your career goals;
- **Work Experience**, listed in reverse chronological order. Include the job title, company name, dates, and a list of responsibilities and achievements;
- **Education**: your academic qualifications, including the name of institutions and dates attended.
- **Skills**: a mix of hard skills (coding, speaking more languages...) and soft skills (fast learner, know how to listen or work in a team...);
- **Certification and hobbies**: you can also mention your certifications (if you have any) or hobbies if they are valuable traits.

Cover Letter

Cover Letter & Job Interview

What is a Cover Letter

A cover letter, or letter of application, introduces you to the employer and points out specifically why you would be suitable for the job. It accompanies the CV; in fact, if it is unconvincing, the employer may not read your CV at all.

The cover letter tells your story and why you want the job.

How it is structured

The **first/opening paragraph** expresses the job you are applying for and how you found out about it. You should also mention your general qualification for the job.

The **middle paragraph** gives more information about why you're suitable for the job. Refer to the CV and choose the most relevant qualification listed there. Discuss them and prove how both your educational and work or training background makes you qualify for the job.

The **final paragraph** expresses your willingness to come for an interview or supply further information. Conclude by saying that you look forward to receiving a reply.

Job Interview

The job interview is a formal meeting (always more often an online meeting) where the employer understands if you're the right fit for the role and company, and where you understand if the company fits you for real.

It is very important, at this point, to make a good impression. Come prepared, maybe by researching the company history, re-read the job description and prepare answers to common questions. It is also very important to be dressed appropriately, maintain a good eye-contact and arrive a few minutes early.

Some of the most requested questions are:

- Why do you want this job?
- What do you think your key strengths are?
- What are your weaknesses?
- Why did you leave your last job?
- Tell me more about a difficult situation you have experienced - how did you deal with it?
- What are your career goals?
- What do you know about our company?

A great technique for answering situational questions (like “Tell me about a time you faced a challenge”) is the STAR one:

- Situation: set the scene;
- Task: describe what needed to be done;
- Action: explain what you did;
- Result: share the positive outcome.

Online Dangers

Online Dangers

Some of them

Phishing

Phishing is a technique which aims to steal personal information such as your passwords or bank account data. The “phishers” hide behind an official entity like banks, streaming services or other companies and send you an email or an SMS that tries to convince you to click on a link, which is a fake link designed to steal your information.

Scamming

Scamming is a general term that refers to all the online frauds that steal your money. An example could be a fake shop with very cheap prices that induces you to buy something from it, stealing your credit card information.

Cyberbullying

Cyberbullying is a form of aggressive behavior carried out repeatedly through digital media (social media, gaming chats, messaging apps).

It goes far beyond the simple insult. Often it includes sharing private photos, isolating or hurting someone or spreading false rumors.

It's a form of bullying that never stops, even if you're alone at home.

Malware

Malware is a type of malicious software made by others to spy, steal or even take control of your device. Malware like Trojans or spyware are made specifically to spy secretly on what we are doing on our device. They can be used to steal information like passwords and emails or take control of our device by creating a backdoor.

How to defend ourselves from online dangers

To stay protected online we have to always be alert.

When we receive a suspicious message, we first of all have to check the sender. Serious companies never ask for sensitive data or passwords via email or SMS.

Never click on suspicious links. Navigate directly to the official website by typing the URL into the search bar.

Avoid sharing your personal information such as your home address or your movements in real-time.

In case of cyberbullying, do not respond to the provocations and block and report the bully. If it begins to become serious, talk to a trusted adult or a teacher.

Speaking of malware, we have to download stuff only from verified sites and, if possible, we could scan our device monthly with an antivirus or use a VPN to protect our data stream online.

How to create a safe password

A strong password is the first line of defense for our accounts.

It should be at least 12-16 characters long.

Combining uppercase, lowercase, special characters and numbers is the best practice to create a secure and solid password.

A thing that we have to avoid is using the same password for multiple accounts or using a password that contains some personal information.

We can also activate the Two-Factor Authentication, so, even if someone finds out your password, they can't access your account without the code that is sent to your email/phone number. This also allows you to check if someone is trying to access your account.

Database

Database

What is

A database is a way of storing a big amount of data so that it can be searched, to find specific records, and processed, to yield (return) specific information.

Databases have become essential in many different areas of modern life, from government to science and education.

A simple database is composed of vertical columns called “fields”: single information like name, surname, nationality...

The horizontal rows are called “records”. The first column (field) is called “key field” because it is unique for each record. The key field is often the primary key of a record: the key which that record is identified.

Type of database

There are three types of database:

Flat file

It consists of one “table”. Records are ordered one after the other, and placed together in a single file. When stored, each value is separated with a comma (;). This is the origin of CSV (Comma Separated Value - a type of file that is used to store data).

Relational

Data is stored in a series of connected tables. The tables are connected by the relation between primary and foreign keys. A foreign key is a “pointer” to a primary key of another table that links them. This makes the database more efficient in terms of storage. It is also easier to manipulate data using a programming language such as SQL.

Non-relational or NoSQL

Data is stored in a much less pre-defined way. They still use tables, rows and columns, but each may vary in the type of data they hold. This database format became very popular in the early 2000s to meet the demand of the companies interested in user-generated content. They are less complex and faster to process.

Data Quality e Consistency

Data redundancy is the unnecessary duplication of data. The relational database removes this problem by dividing them in more tables.

Data inconsistency is the consequence of the redundancy. If the same data is stored in two different places and it is updated in only one of them, the data will be inconsistent.

Data integrity means that data are precise, complete and reliable. DBMS guarantees it through rules (like you can't add a date of birth in the future).

DBMS

A DBMS is a Database Management System: it can store, edit and organize all the data in one central location. It also allows data to be shared and controls who can access them. A DBMS provides a user interface to work with data and automatically creates backup of data.

Furthermore, a DBMS offers **security**, because the data is stored in a central location and different access rights can be given to different people, **efficiency**, it avoids data duplication and inconsistency within the data collection, and **customization**, because access data can be customized to suit the needs of individual users.

DBA

DBA stands for DataBase Administrator.

He is the person responsible for managing , securing and maintaining a database system.

He can decide who can do what with which part of the data.

He installs and configures the programs needed to make the database work.

He is responsible for the backup and the recovery in case of hacker's attack or blackout.

The Fourth Industrial Revolution

The Fourth Industrial Revolution

We have entered a technological revolution that will fundamentally alter the way we live, work and interact.

The First Industrial Revolution used water and steam power to mechanise production. The Second Industrial Revolution used electricity to drive mass production. The Third used electronics and information technology to automate production. Now, the Fourth Revolution, also known as “Industry 4.0” is building on the Third one: the digital revolution. It is characterised by a fusion of technologies which is blurring lines between the physical, digital and biological spheres.

Some of the distinctive characteristics of this Fourth Revolution are:

The incredible **speed** of current discoveries has no historical precedent. The ever increasing advancement of mobile networks contributes to accelerate this evolution.

The **range** of this evolution is enormous. The possibilities of billions of people connected by mobile devices, with unprecedented processing power, storage capacity and access to information are unlimited. And these possibilities will be multiplied by emerging technological discoveries such as 3D printing, artificial intelligence, autonomous vehicles, robotics, biotechnology or Internet of Things (IoT).

A milestone for AI

In 2016, a computer program called AlphaGo beat Lee Sedol, a master player of Go: it was a milestone for AI. Until then, most experts had long agreed that beating a Go champion was far beyond the ability of current AI. This is because Go is more difficult than chess and has more possible positions than atoms in the universe.

AlphaGo, developed by DeepMind company, was designed to adopt a human-like style of gameplay. It followed the “deep learning” technique, which allows a computer to imitate the human-brain and improve and teach itself.

DeepMind designer taught AlphaGo to play the game by feeding it data based on typical Go moves. Then AlphaGo played against itself millions of times, learning from its mistakes and improving quickly.

With deep learning, the program may find a better way to do things than humans have always done.

Will technology replace humans?

As technology advances, many people worry that humans will become redundant. The short answer is no, but the world of work will change completely. Technology will not replace humans entirely, instead, it will change the types of jobs we do.

AI and robots are great at doing repetitive tasks, analyzing big data and working fast. However, they lack uniquely human qualities. Technology cannot replicate:

- **Empathy and Emotion:** jobs like nursing, teaching and therapy need real human connection;
 - **True Creativity:** AI can copy existing ideas, but it cannot think outside the box or feel genuine inspiration;
 - **Critical Thinking:** humans are still needed to make ethical decisions and solve complex, unexpected problems.
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Augmented reality

Augmented reality simply means overlaying digital information onto the physical world via an electronic device, such as mobile phone, tablet or smart glasses.

It enables us to interact with a world that is increased by multimedia and with AR. Unlike virtual reality where we are completely immersed, we can still see what's going on around us.

The ability to stay focused on what is in front of you, without having to look away from the work that you are doing, is one of the key advantages of smart glasses in an industrial context. Software can be integrated with the glasses, for example, to help technicians do repair and maintenance work. The glasses recognise a piece of equipment and show the relevant procedures to make.

There are also augmented reality applications for cheaper devices. Companies such as IKEA use AR to allow users to scan their room at home and then drag and drop furniture into position, using the phone as a viewer.

James Joyce

James Joyce

Life

Joyce was born in Dublin in 1882. However, he spent most of his adult life living in European cities like Trieste, Paris and Zurich.

He had a complicated relationship with Ireland. He left because he felt the country was too traditional and restrictive, but he only wrote about Dublin in all of his books. He remembered every street, pub and detail of his hometown.

“Ulysses”

Published in 1922, Ulysses is considered one of the most important novels in history. The entire book takes place in just one single day and follows a man named Leopold Bloom as he walks around Dublin.

The stream of consciousness

It is a style of writing that tries to show exactly how human thought works. Our minds don't always think in perfect, logical sentences: we jump from one idea to another, mix memories with what we see and get distracted.

Joyce removes normal punctuation like periods and commas. He writes down the thoughts of his characters exactly as they happen.